



ELEC5516

Lecture 12 A

In fiber Grating Optic Sensors

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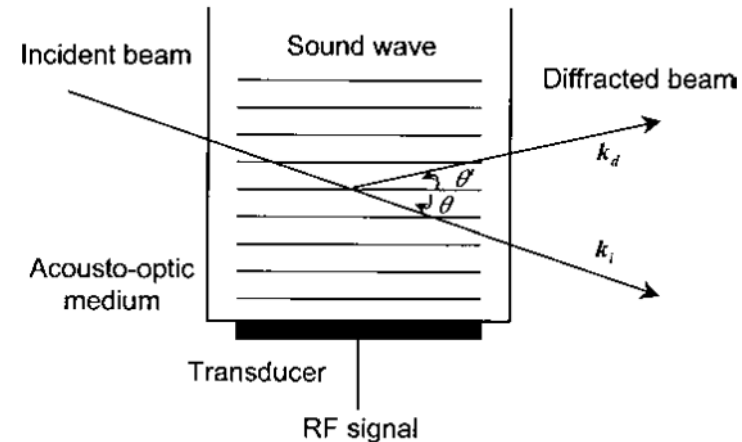
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- › Fiber Bragg Grating (FBG) theory
- › Sensing principle of FBG
- › **Interrogation techniques**
 - Wavelength-Amplitude conversion
 - **Wavelength-Frequency conversion**
 - Wavelength-Position conversion
 - Wavelength-Time conversion
 - Wavelength-Phase conversion
- › Cross sensitivity
- › FBG sensor multiplexing techniques

› Acousto-Optic Tunable Filter Interrogator

- The **acousto-optic interaction** between a light and an **acoustic wave** in a photoelastic medium exhibits a **narrow bandwidth filtering**
- Strong acousto-optic interactions occur when the **Bragg condition** between a **light wave** and a **sound wave** is satisfied
- Varying the acoustic frequency $\rightarrow$ the Bragg condition can be modified $\rightarrow$ **can select the wavelength** of the diffracted light
- The acousto-optic tunable filter (AOTF): varying the frequency of the RF signal $\rightarrow$ **wavelength tuning**
- By changing the RF frequency of the AOTF, it is possible to interrogate a sensor grating.



RF signal
generates the
sound wave via
the transducer



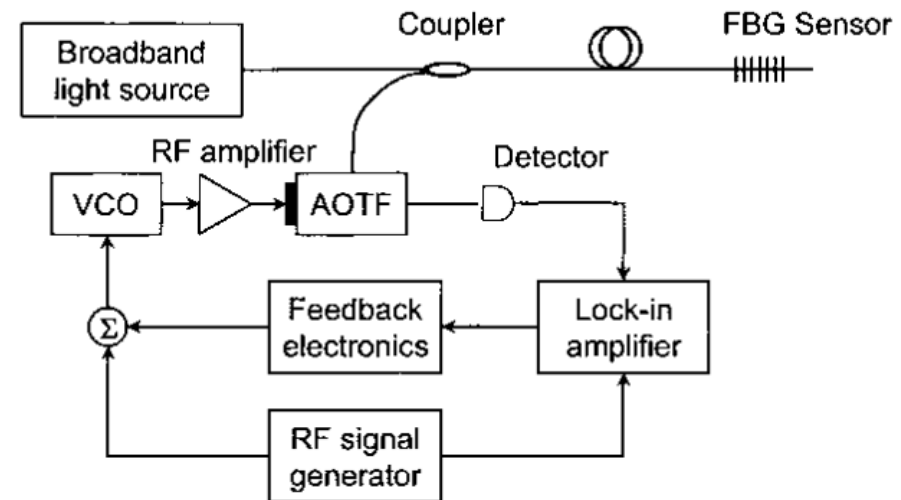
Interrogation Techniques – Wavelength-Frequency conversion

- Acousto-Optic Tunable Filter Interrogator is shown

- Tuning the AOTF by **tuning the VCO** (Voltage controlled oscillator)

- If the **AOTF center wavelength matches** with the FBG center wavelength, some output power will be detected

- If the AOTF is **detuned** away from the FBG wavelength, the electronic feedback system will help modifying the RF frequency

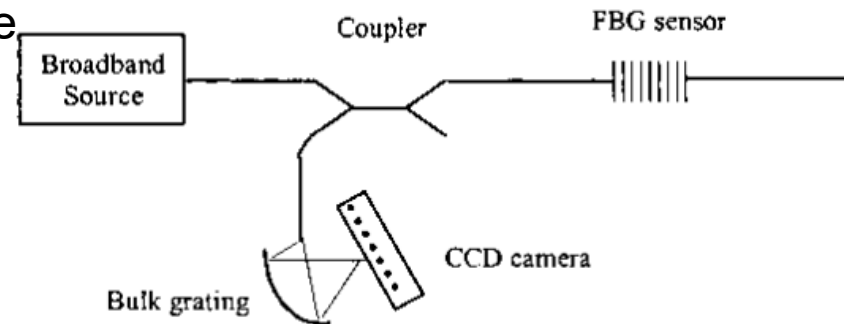
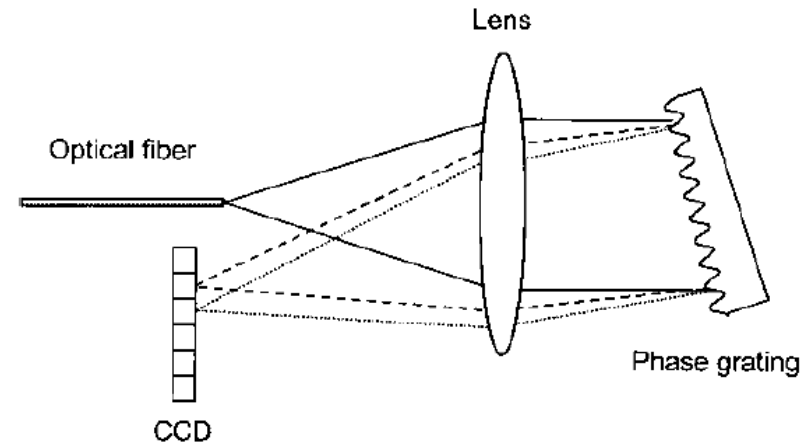


M. G. Xu, H. Geiger, and J. P. Dakin, Modeling and performance analysis of a fiber Bragg grating interrogation system using an acousto-optic tunable filter, *J. Lightwave Technol.*, **14**, pp. 391–396, 1996.

- From the known RF frequency $\rightarrow$ the AOTF center wavelength = the FBG wavelength
- **Temperature Instability: Temperature-tuning coefficient** was 0.03 nm/°C, that is, a temperature change of 1°C in the AOTF would be equivalent to a strain change of 25 $\mu\epsilon$ or a temperature change of 2.3°C
- For the best performance of the AOTF, it must be **temperature-stabilized** or temperature-compensated

› Use fixed **dispersive element**:

- **Distributes different wavelength** components at **different positions** along a line
- A special **phase grating** that diffract light, at different wavelengths, to a **different positions** on a Charge Coupled Device (CCD) camera
- Change in the light wavelength results in the shift of the **light at the detector array** of the CCD
- Resolution depends on the CCD number of pixels and the resolution of the dispersive element
- A strain sensitivity below $1 \mu\epsilon$ was reported



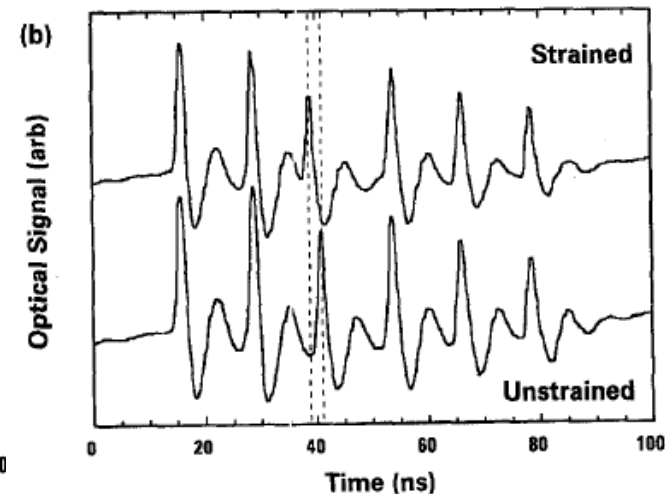
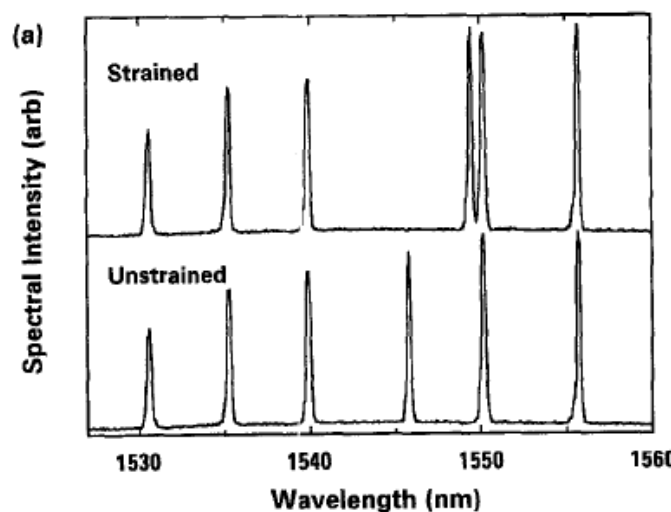
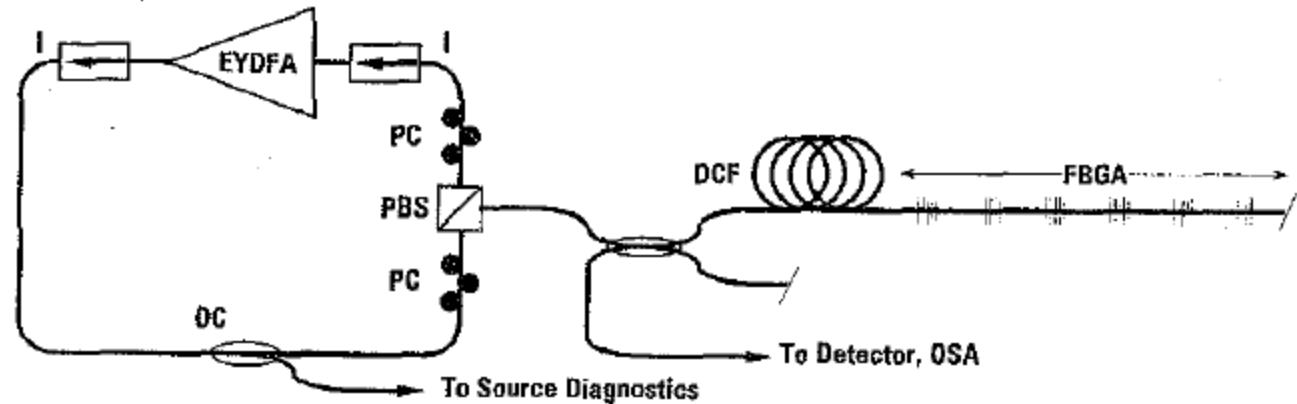
A. D. Kersey, M. A. Davis, H. J. Patrick, M. LeBlanc, K. P. Koo, C. G. Askins, M. A. Putnam, and E. J. Friebele, Fiber grating sensors, *J. Lightwave Technol.*, **15**, 8, pp. 1442–1463, 1997.



Interrogation Techniques – Wavelength-Time conversion

› Using dispersive optical delay line:

- Dispersion compensating fiber (DCF) can have large dispersion parameter
- Ultra fast pulse are launched to FBG sensor via DCF
- Reflection from FBG propagates through DCF and is monitored by fast detector and sampling oscilloscope
- The high dispersion of DCF converts wavelength shift → shift in the pulse arrival time



M. L. Dennis, M. A. Putnam, J. U. King, T.-E. Tsai, I. N. Duling III, and E. J. Friebele, Grating sensor array demodulation by use of a passively mode-locked fiber laser, *Opt. Lett.*, **22**, 17, pp. 1362–1364, 1997.

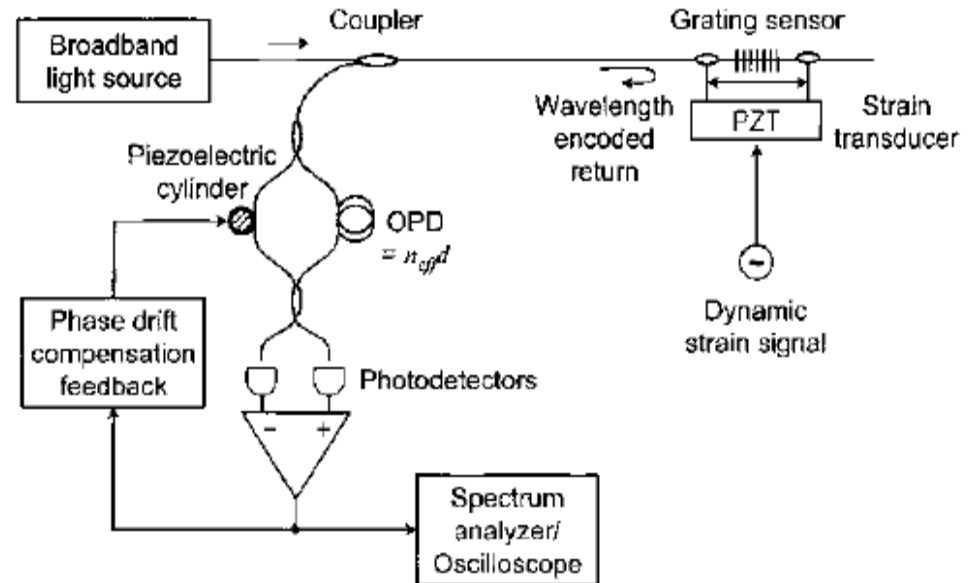
Interrogation Techniques – Wavelength-Phase conversion

- › Using an unbalanced Mach–Zehnder interferometer (MZI) demodulator
- › → the Bragg wavelength change is converted to the phase variation of the interference signal at the detector
- › The light intensity at the detector

$$I = I_0 \left[1 + a \cos \left(2\pi \frac{n_{\text{eff}} d}{\lambda} \right) \right]$$

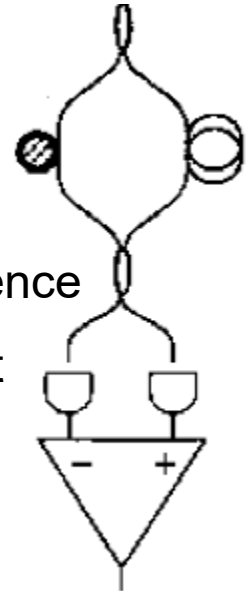
where I_0 is a constant, a is related to the temporal coherence of the light reflected by the FBG, and λ is the free-space wavelength of the light.

- › Bragg wavelength is changed, then the phase in above Eq. is changed
- › By analyzing the phase change, the applied measurand information can be obtained



A. D. Kersey, T. A. Berkoff, and W. W. Morey, High-resolution fibre-grating based strain sensor with interferometric wavelength-shift detection, *Electronics Lett.*, 28, 3, pp. 236–238, 1992.

- › Using broadband sources, the signal level at the photodetector is very low due to the small spectral power density
- › Balanced detection technique helps increase the SNR:
 - lights that cross-coupled the coupler experience a 90° phase change
 - Between two signals at the photodetectors there is 180° phase difference
 - By subtracting, the signal level is doubled and the phase-independent noises are cancelled out
- › Advantages:
 - By controlling the path length difference, the system sensitivity can be easily increased or decreased
 - However, longer the path length difference, the smaller the interference signal becomes
 - Condition for maximum sensitivity: $n_{\text{eff}} d\Delta k = 2.355 \Delta k$ the bandwidth of FBG



- › Fiber Bragg Grating (FBG) theory
- › Sensing principle of FBG
- › Interrogation techniques
- › **Cross sensitivity**
- › FBG sensor multiplexing techniques

- › FBG responses to both strain and temperature → any temperature variation along the fiber is indistinguishable from the strain
- › For static strain measurement, this is a serious issue
- › To recover temperature and strain without ambiguity, a two-grating sensing system is used

$$\begin{pmatrix} \Delta\lambda_1 \\ \Delta\lambda_2 \end{pmatrix} = \begin{pmatrix} K_{1\varepsilon} & K_{1T} \\ K_{2\varepsilon} & K_{2T} \end{pmatrix} \begin{pmatrix} \varepsilon \\ T \end{pmatrix}$$

where $\Delta\lambda_1$ and $\Delta\lambda_2$ are the wavelength shifts from two grating sensors, respectively, $K_{1\varepsilon}$ and K_{1T} represent the response to strain and temperature for grating 1, $K_{2\varepsilon}$ and K_{2T} for grating 2.

- › To avoid ambiguity:
 - $K_{1\varepsilon} = K_{2T} = 0$, or $K_{2\varepsilon} = K_{1T} = 0$
 - One element in the response matrix being zero
 - The two sensors have very different responses so that the determinant of the response matrix is nonzero

Cross-Sensitivity – Reference Grating

- › Using another FBG that is shielded from strain/temperature and measures only one perturbation $K_{IT} = 0$
 - A strain-free temperature reference grating that is co-located with the strain sensors, experiencing the same thermal environment
 - Temperature compensation is obtained by subtracting the wavelength shift induced by temperature
 - Cost extra gratings and their associated interrogation units
- › Passive temperature-compensating package to nullify the temperature effect:
 - grating is mounted under tension in a package comprised of two materials of silica and aluminum, which have different thermal expansion coefficients
 - As the temperature rises the strain on the grating is released just enough to cancel the thermal shift in Bragg wavelength

Cross-Sensitivity – Two FBGs with very different responses

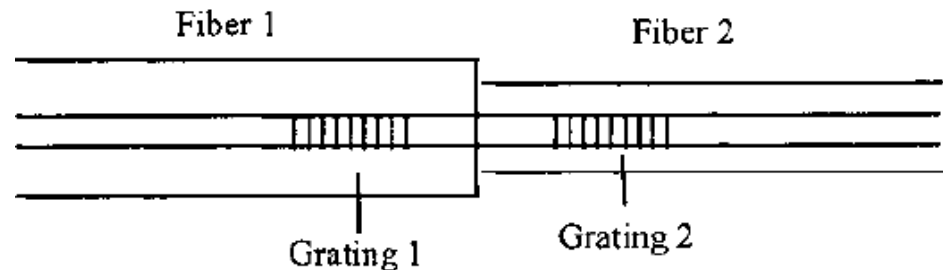
- › The elimination of cross-sensitivity can be achieved by measurements at two different wavelengths or two different optical modes, for which the strain and temperature responsivity is different
- › The photoelastic and thermo-optic coefficients of the fiber glass are wavelength dependent
- › There is a small variation in the ratio of responses of FBGs written in 848 nm and 1298 nm
- › This leads to a nonzero determinant of response matrix K

$$\begin{pmatrix} \Delta\lambda_1 \\ \Delta\lambda_2 \end{pmatrix} = \begin{pmatrix} K_{1\varepsilon} & K_{1T} \\ K_{2\varepsilon} & K_{2T} \end{pmatrix} \begin{pmatrix} \varepsilon \\ T \end{pmatrix}$$

- › Changes in both strain and temperature can be determined by solving the Eq.

Cross-Sensitivity – FBGs with different fiber diameters

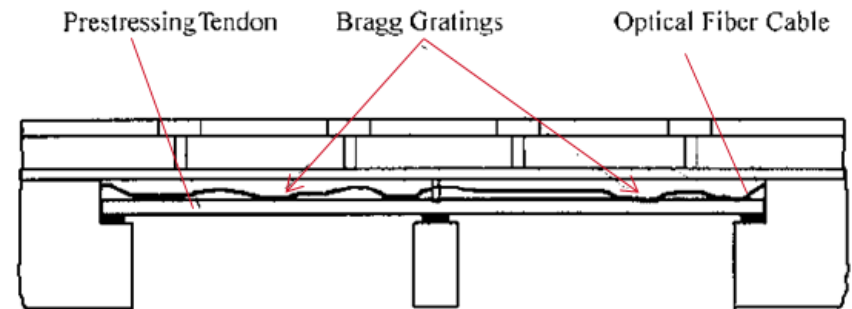
- › Fiber with a larger diameter experiences smaller strain than that with a smaller diameter
 - A change in temperature will shift the center wavelengths of the two gratings by similar amounts
 - A change in strain results in the center wavelengths' moving by different amounts
- › Relative wavelengths provide a measurement of the strain
- › Absolute wavelength shifts include information on both the temperature and strain



- › Fiber Bragg Grating (FBG) theory
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- › **FBG sensor multiplexing techniques**

FBG sensor multiplexing techniques

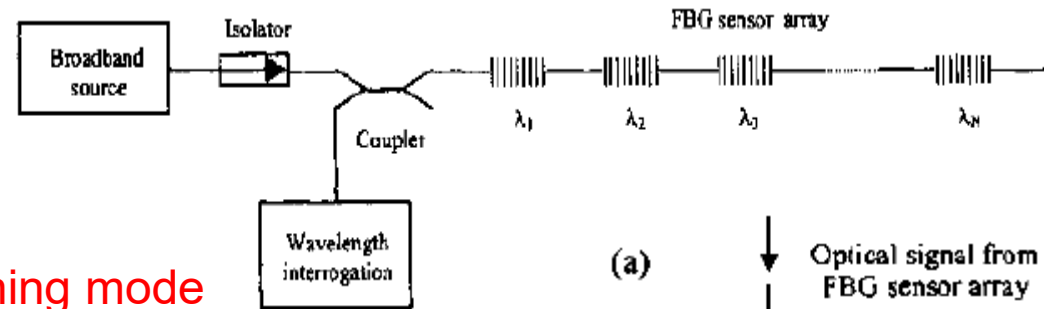
- › In many applications a **large number of sensors needed** to achieve a distributed measurement of the parameters
- › Example: Strain monitoring a bridge



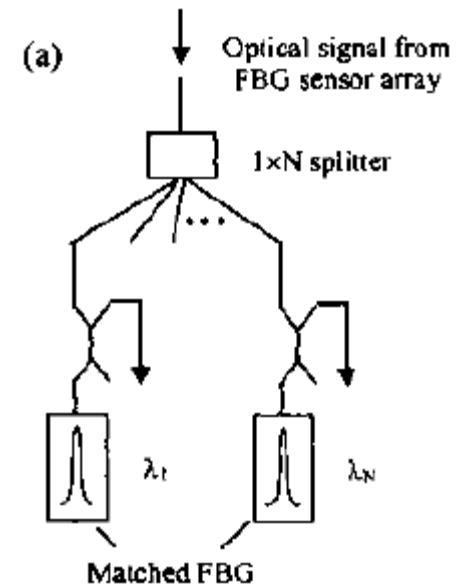
- › Advantages of FBG:
 - **Many** sensors can be **interrogated through a single fiber**
- › Common multiplexing techniques:
 - Wavelength division multiplexing (WDM)
 - Time division multiplexing (TDM)
 - WDM/TDM

FBG sensor multiplexing techniques - WDM

- › WDM utilizing the wavelength-encoding feature of an FBG-based sensor
- › **Physical spacing** between FBG sensors can **be as short as desired** to give accurate distributed information

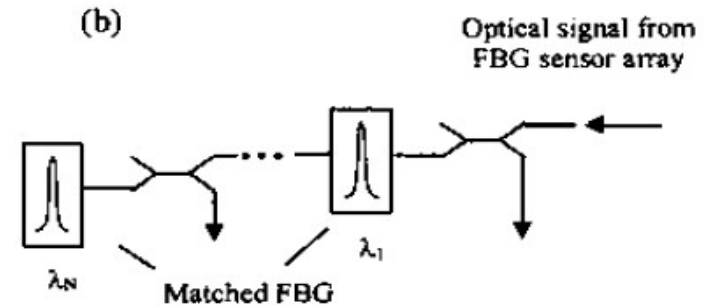


- › Basic WDM configuration →
 - Tunable Fabry–Perot filter in **scanning mode** was used to distinguish between the wavelength shifts
- › A parallel topology is used to allow **simultaneous** interrogation:
 - More FBG sensors lead to a **larger power penalty**

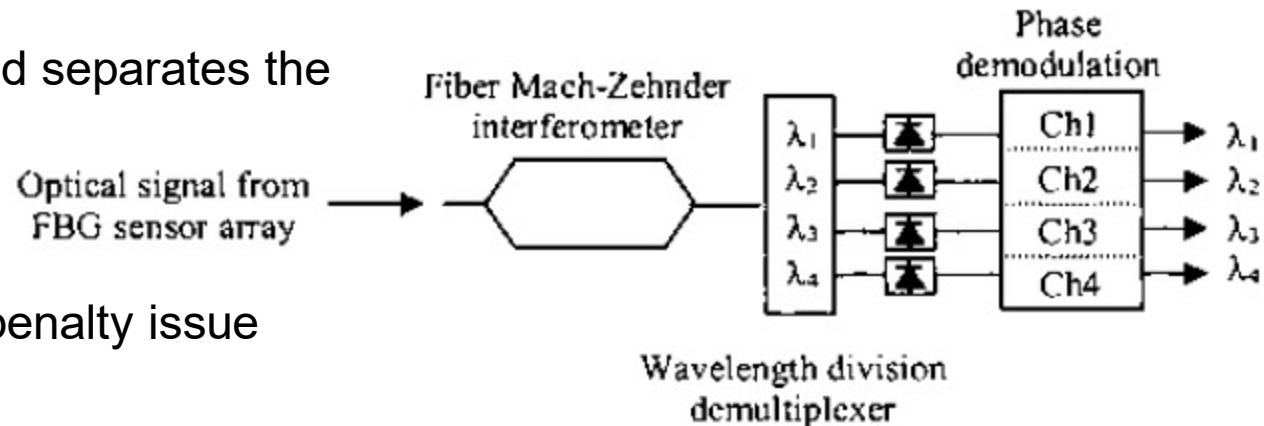


FBG sensor multiplexing techniques - WDM

- › An improved scheme using a serial matched FBG array is reported

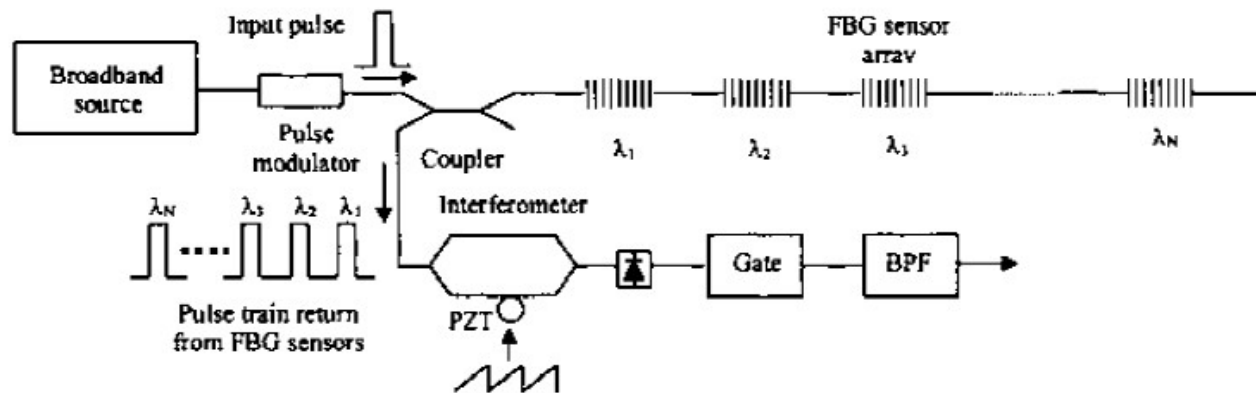


- › WDM using bandpass wavelength division demultiplexer
 - Reflected optical signal from the FBG array fed to a MZI which is a wavelength shift detector
 - A wavelength splitter and separates the composite signal into discrete channels
 - Overcomes the power penalty issue



FBG sensor multiplexing techniques - TDM

- › In TDM, **optical pulses** are employed
- › Basic TDM configuration is shown

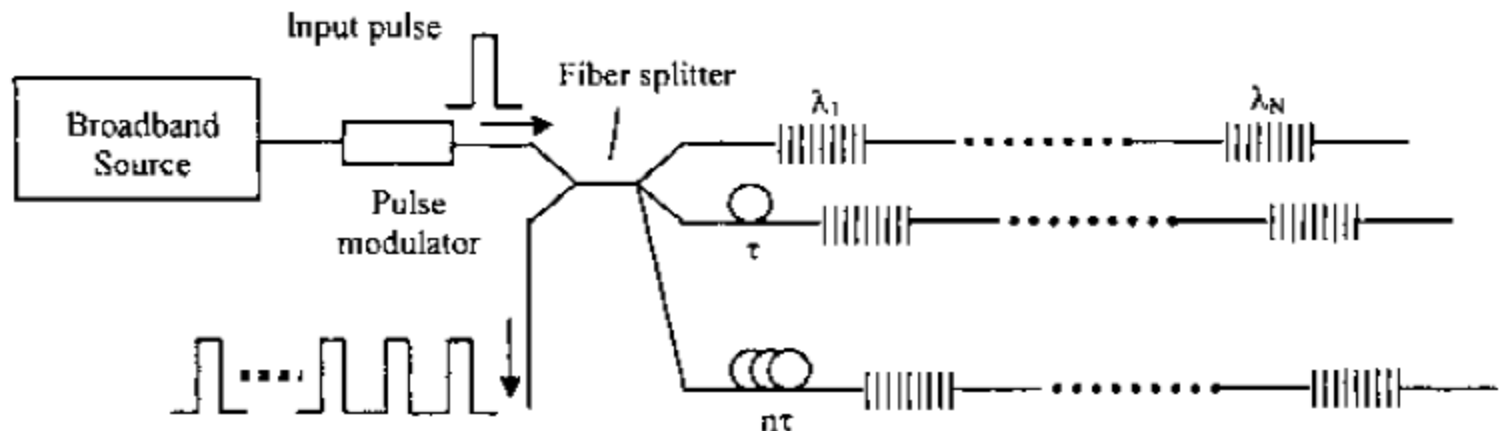


- › After photodetection, the return pulses between adjacent FBGs are recovered, with the **two pulses separated by the round-trip time** between gratings
- › The input **pulse width < the round-trip time** between two gratings
- › **Number of multiplexed FBG sensors** (or the density of FBG sensors) depends on the **pulse width** → can be improved by high-speed gating device



FBG sensor multiplexing techniques – WDM/TDM

- › WDM provides for tens of FBG sensors per fiber
- › TDM can **multiply** this number several times by reusing the wavelength spectrum of the source
- › A parallel mixed WDM/TDM configuration is shown
 - A certain length of **fiber delay line separates the time window** for each WDM set



- › **Introduction**
- › Sensing principle of polarimetric OFS
- › Interrogation/Demodulation techniques
- › Cross sensitivity
- › Coherence multiplexed sensors
- › Optical current measurement
- › Optical voltage sensor

› Polarimetric Sensors:

- Apply polarimetric measurement techniques, which **relates** the induced **polarization change** of the light within an **optical fiber** to the **measurands** of interest.

› Advantages:

- Passive, lightweight
- High bandwidth
- Cost-effective

› Applications:

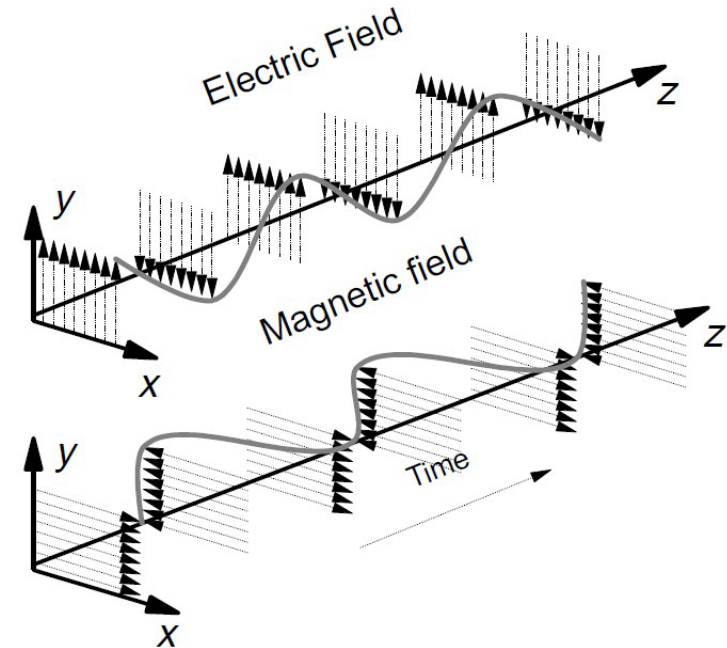
- Strain and Temperature sensors
- Fiber Optic Current Sensors for high voltage applications
- Etc.

- › Introduction
- › **Sensing principle of polarimetric OFS**
- › Interrogation/Demodulation techniques
- › Cross sensitivity
- › Coherence multiplexed sensors
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- › Optical voltage sensor

Sensing principle of polarimetric OFS

- › How to induce a change in polarization state of the light?
 - Light is made of EM field, electric field and magnetic field are perpendicular
 - Looking at the example on the right, if considering the electric field only, we can see that electric field lies only on the y-axis
 - The light in this case is said to be linearly polarized
 - Assuming now there is another light with the same wavelength but the electric field is orthogonal to the first light:
 - Depending on the phase lag between two lightwaves, the combined lightwave will have different states of polarization

Polarization state changing of a lightwave



Sensing principle of polarimetric OFS

- › Polarization change is all about **optical phase change**!
- › Phase of a lightwave guided by a fiber of length L

$$\phi = \beta L = n_{\text{eff}} k_0 L$$

β is the propagation constant of the mode
 n_{eff} is its effective index.

k_0 is the free-space wave number of the light equal to $2\pi/\lambda_0$

- › For any two modes guided by the fiber, the **difference in their optical phases** after length L can be expressed

$$\Delta\phi = \Delta\beta L = \Delta n_{\text{eff}} k_0 L$$

Δn_{eff} is the difference in effective indices between the two modes

- › The overall phase response is proportional to the length of the fiber exposed to the external field → consider the response of the sensor in relation to a **unit sensor length**



Sensing principle of polarimetric OFS

- › If the fiber is subjected to an external strain ϵ ,

$$\frac{1}{L} \frac{\partial(\Delta\phi)}{\partial\epsilon} = k \frac{\partial(\Delta n_{\text{eff}})}{\partial n} \frac{\partial n}{\partial\epsilon} + k \frac{\partial(\Delta n_{\text{eff}})}{\partial D} \frac{\partial D}{\partial\epsilon} + k \Delta n_{\text{eff}} \frac{1}{L} \frac{\partial L}{\partial\epsilon}$$

Photo elastic
effect:
mechanical
strain to
refractive index
change ←
dominant

longitudinal
strain on the
diameter of the
fiber to
refractive index
change →
negligible

physical
change in
length due to
strain

$$\frac{\partial n}{\partial\epsilon} = -\frac{n^3}{2} [p_{12} - \nu(p_{11} + p_{12})]$$

ν is Poisson's ratio taken to be the same in both core and cladding
 p_{11}, p_{12} are the strain optic coefficients

- › Given an **optical phase change** → can derive **applied strain**



Sensing principle of polarimetric OFS

- › If the fiber experiences a change in temperature T ,

$$\frac{1}{L} \frac{\partial(\Delta\phi)}{\partial T} = k \frac{\partial(\Delta n_{\text{eff}})}{\partial n} \frac{\partial n}{\partial T} + k \frac{\partial(\Delta n_{\text{eff}})}{\partial L} \frac{\partial L}{\partial T} + k \Delta n_{\text{eff}} \frac{1}{L} \frac{\partial L}{\partial T}$$

the change in
the refractive
index of the fiber
due to the pure
temperature
effect ←
dominant

change in
refractive index
due to the
changes in the
fiber
dimensions via
the photoelastic
effect.

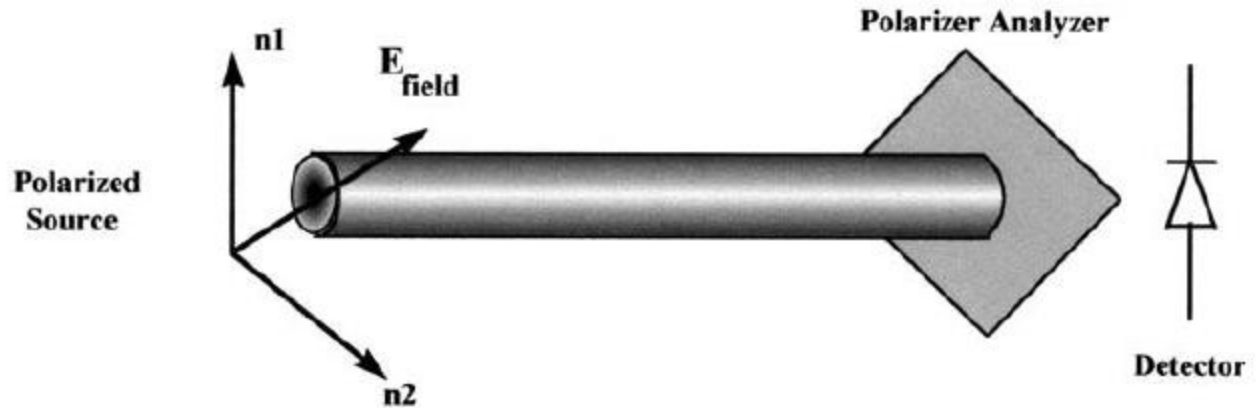
change in fiber
length due to
thermal
expansion

- › Given an optical phase change → can derive the change in temperature

- › Introduction
- › Sensing principle of polarimetric OFS
- › **Interrogation/Demodulation techniques**
- › Cross sensitivity
- › Coherence multiplexed sensors
- › Optical current measurement
- › Optical voltage sensor

Interrogation/Demodulation techniques

› A fiber polarimeter:



- Linearly polarized light is launched at 45° to the principal axes of a fiber
- Output polarizer analyzer oriented at 90° to the input polarization state
- The intensity detected after the polarization analyzer will be

$$I = E^2 = \frac{I_o}{2} [1 + \cos \phi_2]$$

I_o is the total output power
 ϕ_2 is the induced phase retardance

- Change in polarization state is converted into intensity
- Polarization change in fiber depending on stress and heat

- › Introduction
- › Sensing principle of polarimetric OFS
- › Interrogation/Demodulation techniques
- › **Cross sensitivity**
- › Coherence multiplexed sensors
- › Optical current measurement
- › Optical voltage sensor

- › In cases of static strain measurement, some means of recovering the strain and temperature information from a single-sensing fiber is required
- › Generally, the solution is to interrogate the sensor with two different measurement approaches at the same time

$$\begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix} = \begin{bmatrix} K_{1T} & K_{1\varepsilon} \\ K_{2T} & K_{2\varepsilon} \end{bmatrix} \begin{bmatrix} T \\ \varepsilon \end{bmatrix} \quad \Delta = K_{1T}K_{2\varepsilon} - K_{2T}K_{1\varepsilon}$$

- Provided the determinant is not zero, this equation may be inverted and T and ε can be calculated
- One feasible approach is to implement two polarimeters on the same fiber, one on the LP₀₁ mode and other on the LP₁₁.

$$\begin{bmatrix} \delta\Delta\phi_{LP_{01}} \\ \delta\Delta\phi_{LP_{11}} \end{bmatrix} = \begin{bmatrix} -4.87 \times 10^{-3} & -0.882 \\ 7.2 \times 10^{-3} & -0.846 \end{bmatrix} \begin{bmatrix} \delta\varepsilon (\text{rad}/\mu\text{m}) \\ \delta T (\text{rad}/^\circ\text{C m}) \end{bmatrix}$$

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- › **Coherence multiplexed sensors**
- › Optical current measurement
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Coherence multiplexed sensors

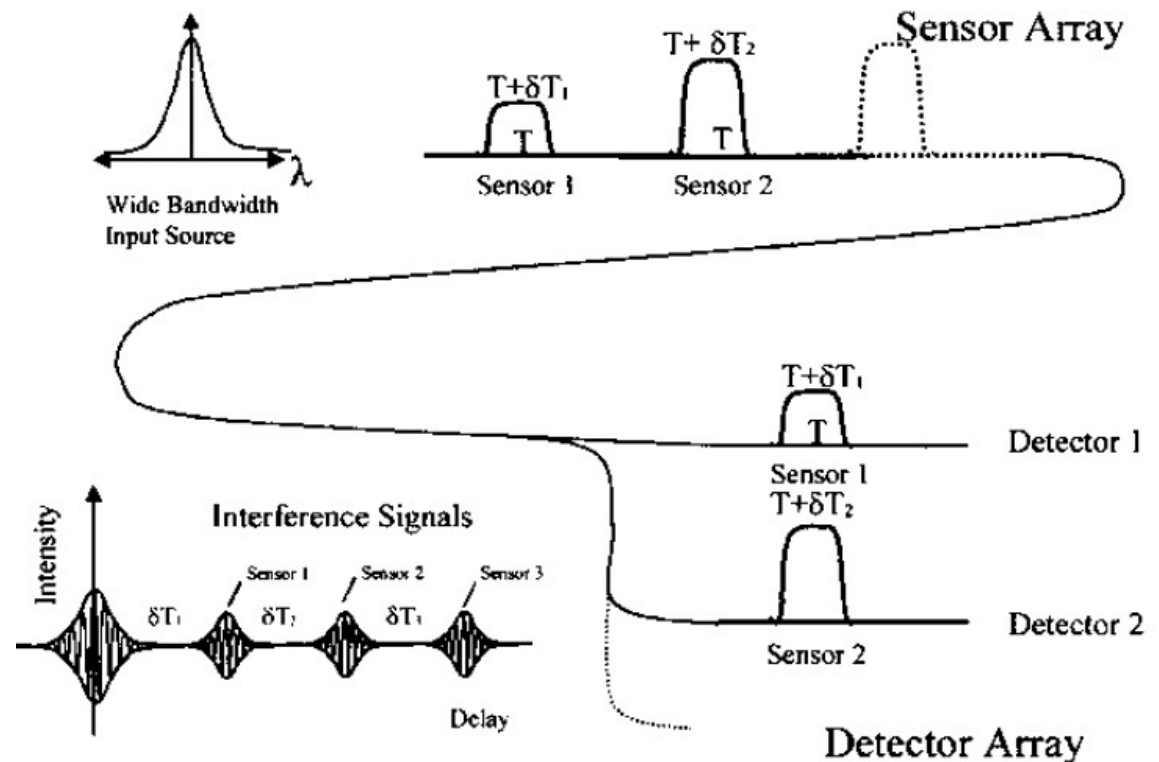
- › TDM is limited by the processing speed of the associated electronics
- › Coherence multiplexing can solve this problem!

-Polarimetric sensor measures the relative phase delay between light traveling in two arms of an interferometer (i.e. two light modes).

-This information is recovered by processing the **interference signal** derived from a polarizer.

- The **interference signal can only be recovered** if the 2 signals are **coherent** – that is the relative **time delay between the 2 modes does not exceed the coherence time** of the light source.

- Coherence multiplexed systems exploit this constraint to give each sensor a uniquely identifiable signature



- › Introduction
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- › Interrogation/Demodulation techniques
- › Cross sensitivity
- › Coherence multiplexed sensors
- › **Optical current measurement**
- › Optical voltage sensor

- › The **optical current transducer** (OCT) determines the current flow in an electrical conductor by measuring the **magnetic flux density** within the **vicinity of the conductor**
- › The **magnetic flux density** is determined by the **polarization rotation** induced as the light propagates through a guiding medium (optical fiber or a sensitive crystal) in the vicinity of the conductor
- › The angular rotation θ experienced by the light passing through the sensor is

$$\theta = VBl$$

B is the magnetic field strength

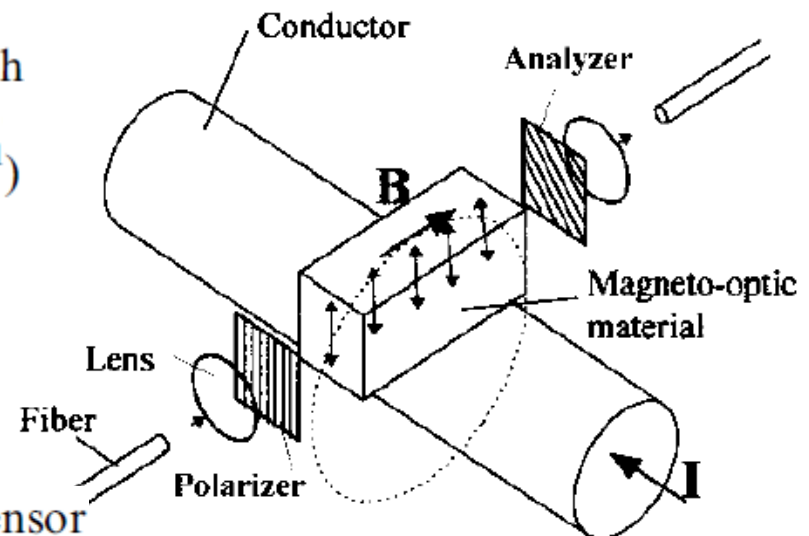
l is the length of the sensor

Verdet constant (V , $\text{rads T}^{-1} \text{m}^{-1}$)

- › The transmitted intensity is

$$I = I_0(1 + \sin(2\theta))/2$$

I_0 is the input intensity of the light delivered to the sensor



Optical current measurement – wound-fiber type OCT

› Wound-fiber type OCT:

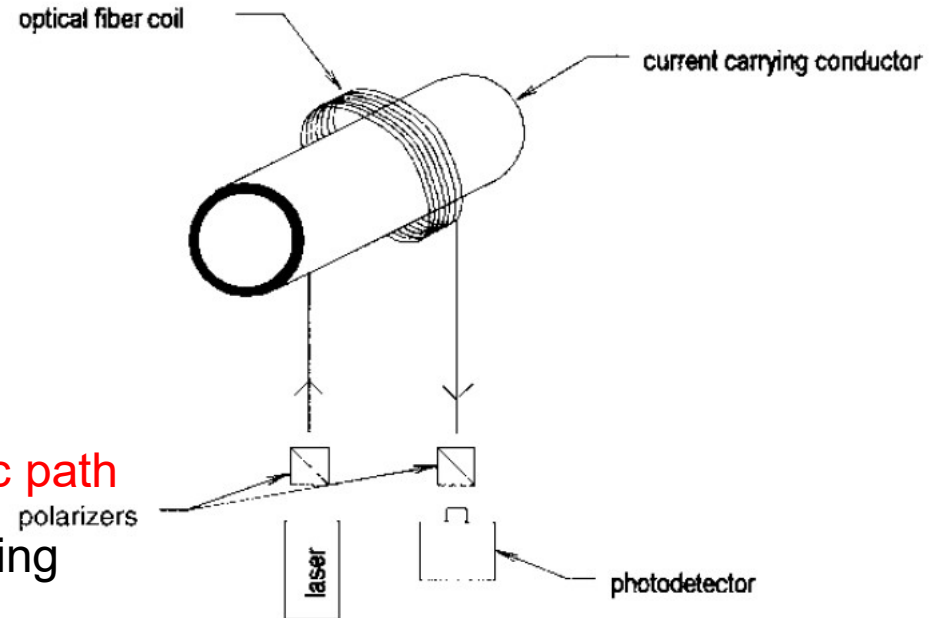
- Optical fiber itself as the medium for detecting the induced magnetic field

› Advantages:

- Simple and cost-effective
- Fiber coil **totally encloses the magnetic path**
→ **isolation** from other potentially corrupting magnetic sources

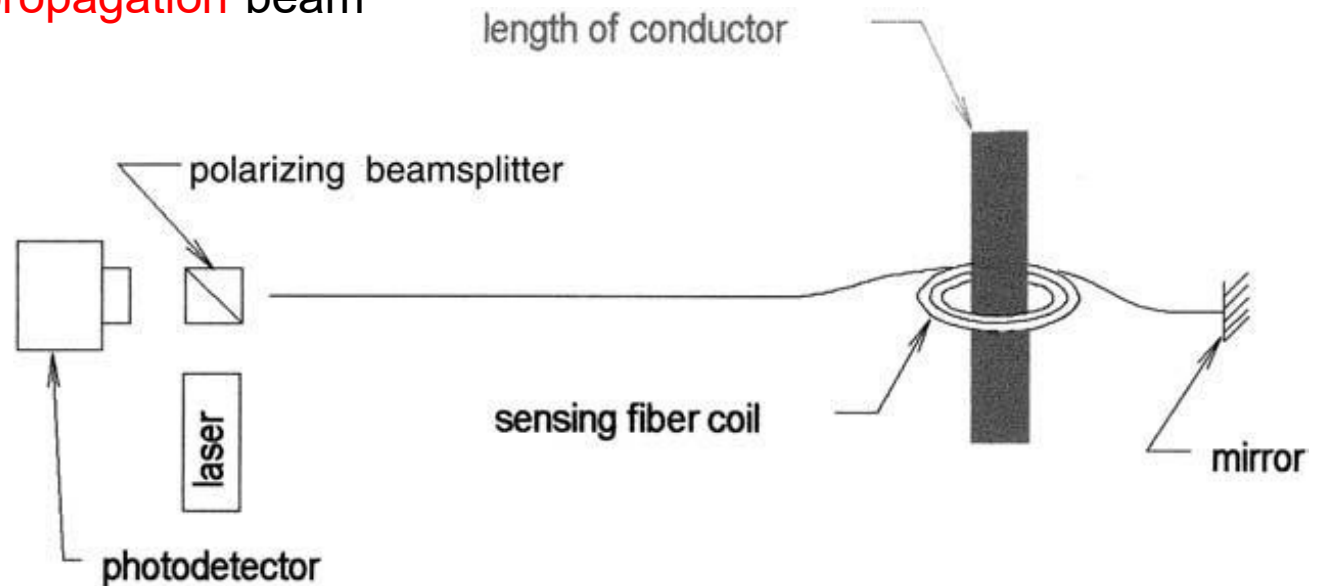
› Drawbacks:

- **Verdet constant** of silica fiber is significantly **lower** than that of the crystalline material
- **Instability from the linear bi-refringence** induced within the fiber coils



› Reflective-type wound-fiber OCT

- **Nonreciprocal** nature of the Faraday effect → **double** the sensing length of fiber
- **Suppress** the influence of **vibration-induced bi-refringence**, influence on the **clockwise-propagation** beam will be in anti-phase to the effect on the **counterclockwise-propagation** beam

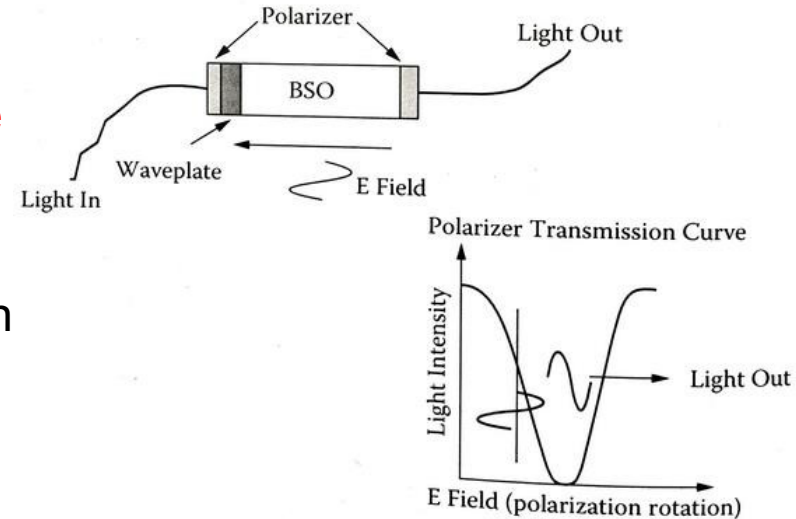


M. Takahashi, H. Noda, K. Terai, S. Ikuta, and Y. Mizutani, Optical current sensor for gas insulated switchgear using silica optical fiber, *IEEE Trans. Power Delivery*, **12**, 4, pp. 1422–1428, Oct. 1997.

- › Introduction
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- › Optical current measurement
- › **Optical voltage sensor**

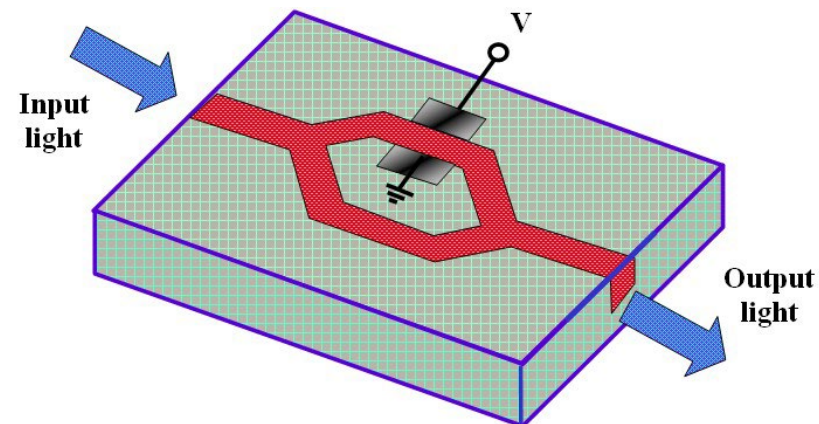
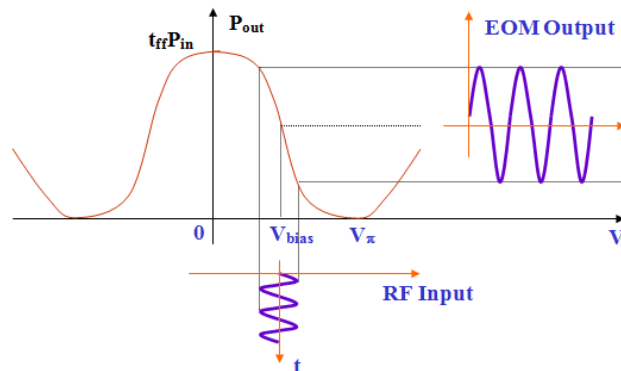
› Polarimetric Voltage Sensors:

- Exploiting the **Pockel effect**: **material refractive index changes** in electric field intensity
- The sensor measures **the polarization rotation** produced in an optically active crystal (BSO) in the presence of an electric field



› Integrated Optic Devices:

- Similar principle of operation to an integrated optical intensity modulator



Capacitive voltage divider in Optical voltage sensor

- › It is **not practical** to let the **entire electric field** drop across the crystal of the voltage sensor
- › Thus, a **simple capacitive divider** is introduced to ensure that the electric field developed across the crystal **remains constant in relation to the line potential**
- › This is **not possible** in the case of **electrical voltage sensors** because their capacitance influences the stability of the divider network.
- › However, an optical voltage sensor equivalent **impedance** can be considered **infinite** → loading effects can be disregarded.

